Short questions

|!|BCESB0210000001|!| ★

Photosynthesis consists of light-dependent and light-independent stages.

(a) Describe how light intensity affects the light-dependent stage. (3 marks)

(b) (i) Suggest ***two*** limiting factors that affect the light-independent stage directly. (2 marks)

(ii) Explain how these factors affect the light-independent stage. (2 marks)

##

(a) The higher the light intensity, the more light energy is absorbed by chlorophylls to excite more of its electrons. (1)
More energy is thus released by the electrons to synthesize more ATP and split more water molecules into hydrogen and oxygen / photolyse water molecules. (1)
More ATP and hydrogen can then be used in the next stage of photosynthesis. (1)

(b) (i) Temperature (1)
Carbon dioxide concentration (1)

(ii) An increase in temperature can increase the activity of the enzymes involved in photosynthetic reactions and hence the rate of photosynthesis. (1)
An increase in carbon dioxide concentration provides more carbon to be fixed into triose phosphate, the primary photosynthetic product. (1)

(7 marks)

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##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)(i)	Applying concepts	2
(b)(ii)	Applying concepts	2

|!|BCESB0210000002|!| ★ **Cross topic**

For each of the organelle structures listed in Column 1, select from Column 2 ***one*** reaction that occurs there. Put the appropriate letter in the space provided. (3 marks)

Column 1		***Column 2***
thylakoid	(a) _____	A. Krebs cycle
chloroplast's stroma	(b) _____	B. photolysis
mitochondrial matrix	(c) _____	C. Calvin cycle
		D. glycolysis
		E. oxidative phosphorylation

##

(a) B (1)
(b) C (1)
(c) A (1)

(3 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

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|!|BCESC0210000003|!|

The following paragraph describes the Calvin cycle. Complete the paragraph with the appropriate term(s). (5 marks)

In the Calvin cycle, a (a) _____ molecule combines with a five-carbon (5-C) carbon dioxide (b) _____ to form an unstable 6-C compound, which immediately splits into two molecules of a 3-C compound. The 3-C compound is reduced by (c) _____ to form another 3-C compound called (d) _____, which can be used to form a 6-C (e)_____ molecule and the 5-C compound.

##

(a) carbon dioxide (1)
(b) acceptor (1)
(c) NADPH (1)
(d) triose phosphate (1)
(e) glucose (1)

(5 marks)

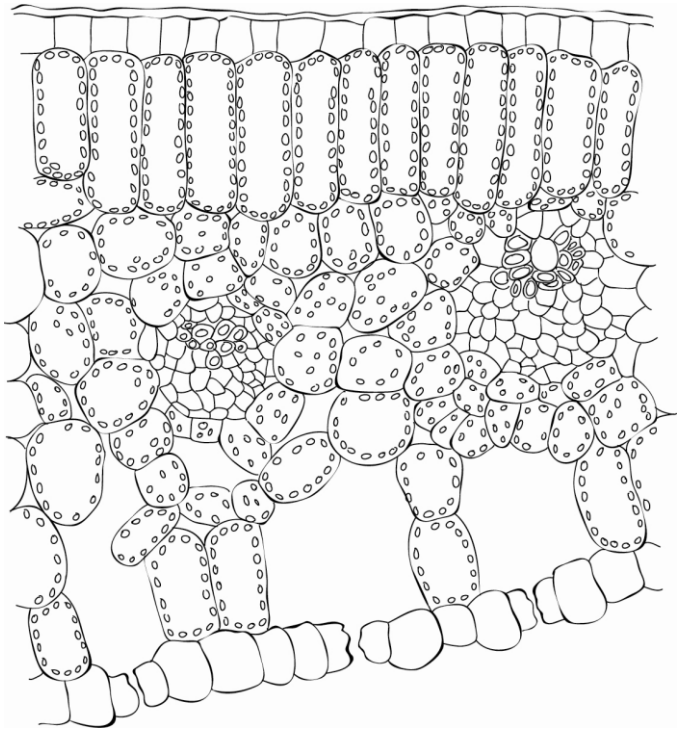
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

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|!|BCESB0210000004|!| ★

The following diagram shows the cross section of a leaf.



According to the diagram, describe how the leaf is adapted to

(a) maximize the rate of photosynthesis. (3 marks)

(b) minimize the rate of water loss. (2 marks)

##

(a) Any ***three*** of the following: (3)

- Palisade mesophyll cells are tightly packed.
- Palisade mesophyll cells are densely packed with chloroplasts.
- Spongy mesophyll cells are loosely packed to facilitate gas exchange.
- Xylem provides water for photosynthesis.
- Stomata allow the exchange of carbon dioxide and oxygen.

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(b) Any ***two*** of the following: (2)

- More stomata are located on the lower surface.
- Epidermal cells are covered with a waxy cuticle.

(Or other reasonable answers)

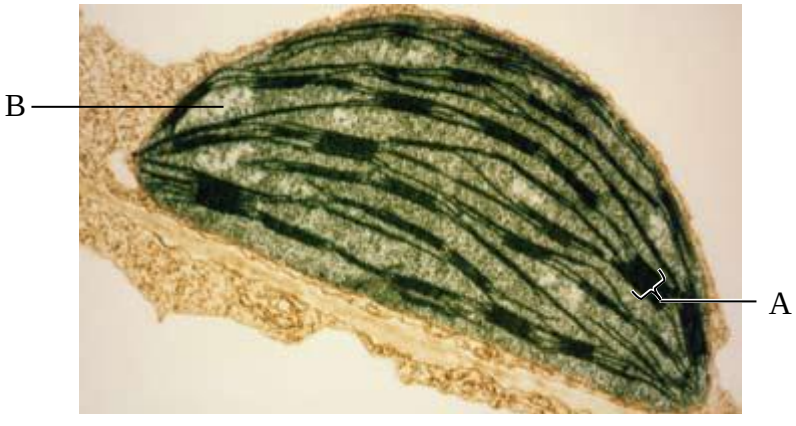
(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Analysing data	2

|!|BCESB0210000005|!| ★

The following electron micrograph shows a chloroplast.



(a) Identify A and B. (2 marks)

(b) Suggest ***two*** ways that chloroplasts are adapted to photosynthesis. (2 marks)

##

(a) A: thylakoids (1)

B: stroma (1)

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(b) Any ***two*** of the following: (2)

- The extensive network of tightly-packed thylakoid membranes provides a large surface area for the attachment of chlorophyll molecules. This allows efficient light absorption.
- The stroma contains enzymes that are involved in the reactions of photosynthesis.
- The grana in thylakoids are interconnected to allow the efficient transport of photosynthetic products.

(Or other reasonable answers)

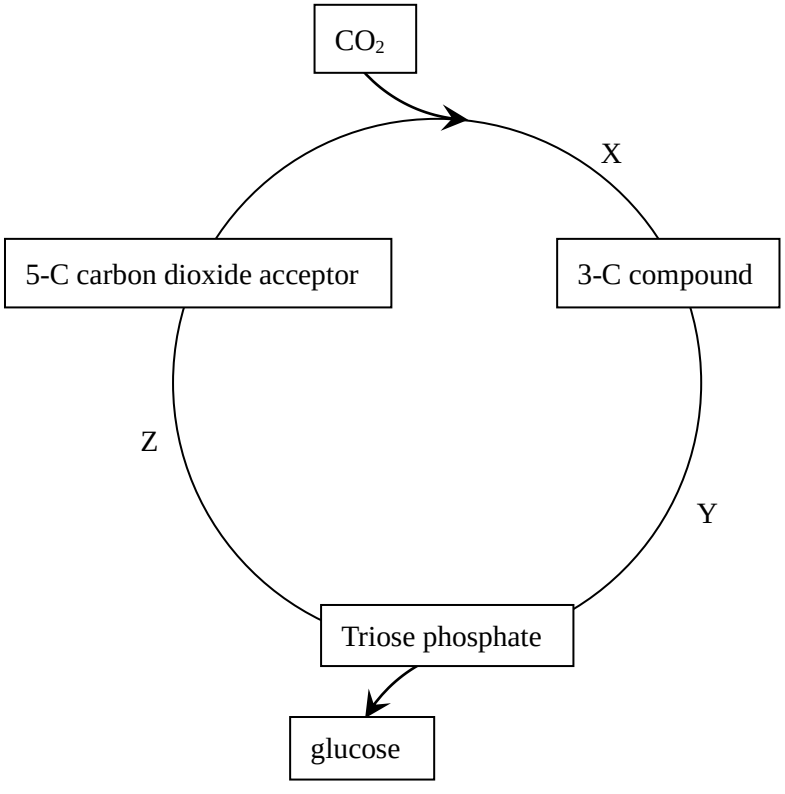
(4 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	2

|!|BCESB0210000006|!| ★

The following diagram shows the light-independent stage of photosynthesis.



(a) Name this light-independent stage of photosynthesis. (1 mark)

_____

(b) The light-independent stage consists of three processes: X, Y and Z.

(i) State the process(s) that requires energy in the form of ATP. (1 mark)

_____

(ii) State the process(s) that requires hydrogen from NADPH. (1 mark)

_____

(c) How many triose phosphate molecules are required to synthesize one molecule of glucose? (1 mark)

_____

##

(a) The Calvin cycle (1)

(b) (i) Y, Z (1)

(ii) Y (1)

(c) Two (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	1
(c)	Understanding basic concepts	1

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|!|BCESB0210000007|!| ★

The table below shows the rates of photosynthesis in an herbaceous plant under various carbon dioxide concentrations.

Carbon dioxide concentration (arbitrary units)	**Rate of photosynthesis (arbitrary units)**
30	20
60	40
90	60
150	80
240	100
300	120
360	120
390	120

(a) Describe and explain the trend shown by the data in the table above. (4 marks)

(b) Besides carbon dioxide concentration, suggest ***two*** other factors that will affect the rate of photosynthesis. (2 marks)

##

(a) The rate of photosynthesis increases along with carbon dioxide concentration. (1)
This is because the plant can obtain more raw materials to carry out photosynthesis. (1)

The rate of photosynthesis levels off when the carbon dioxide concentration reaches and exceeds 300 arbitrary units. (1)
This is because other factors become limiting. (1)

(b) Light intensity (1)
Temperature (1)
(Or other reasonable answers)

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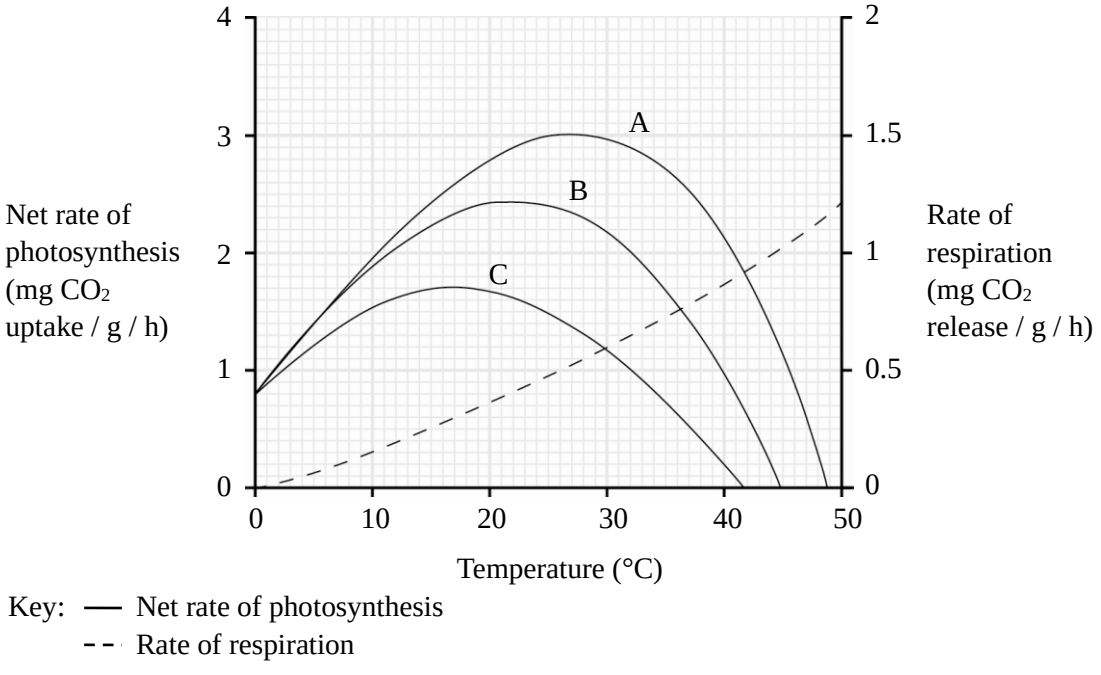
(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	4
(b)	Understanding basic concepts	2

|!|BCESA0210000008|!| ★★ **SI**

Tommy investigated the effect of temperature on the net rate of photosynthesis in a plant species X at three different light intensities. He also investigated the effect of temperature on the rate of respiration in the plant (kept in a dark room). The following graph shows the results. Curves A, B and C show the results under high, medium and low light intensities respectively.



(a) Name ***one*** factor that needed to be controlled in this investigation. (1 mark)

_____

(b) Name ***two*** factors that limited the rate of photosynthesis between 0 °C and 10 °C at low light intensity. (2 marks)

_____

_____

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(c) Plant species X grows in mountain habitats. The area where Tommy collected his plant sample will become much cloudier and warmer during the summer, when temperatures could rise from 20 °C to 25 °C. Explain how this change in weather will affect the growth of plant species X. (3 marks)

##

(a) Any ***one*** of the following: (1)

- Carbon dioxide concentration
- The size of the plant to be used

(Or other reasonable answers)

(b) Temperature (1)

Light intensity (1)

(c) Growth will decrease when the temperature increases from 20 °C to 25 °C under cloudier conditions. (1)

This is because the rate of respiration will increase at higher temperatures while the rate of photosynthesis will decrease as the process is limited by light intensity. (1)

(1)

(6 marks)

##

	Skills assessed	Marks
(a)	Practical skills	1
(b)	Analysing data	2
(c)	Analysing data	3

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